

.NET Developer Assignment – Paint Tint Formula Calculator

Revest

Overview

Build a small end-to-end feature that calculates paint tinting formulas, using a WPF desktop client, an ASP.NET Core Web API and a database.

Item	Detail
Time limit	2 days
Tech stack	.NET 8 or later, WPF (MVVM), ASP.NET Core Web API, EF Core
Database	SQL Server or SQLite
Focus	Clean code and correct business logic, not the number of features

Scenario

A paint store operator tints a can of paint at the counter. The operator picks a shade, a base and a can size. The system shows how much of each colorant to dispense and the price, then saves the dispense record.

Key terms:

- **Base:** the untinted paint the colorant is added to (Pastel, Medium or Deep).
- **Colorant:** a concentrated tint added to the base, such as Black or Oxide Red.
- **Shade:** a named color, such as "Ocean Mist (OM-201)", with a formula of colorant amounts.

Business rules

1. Each shade's formula gives colorant amounts in ml for a **1L can**.
2. Can sizes are 1L, 4L, 10L and 20L. Amounts scale linearly with the can size.
3. Round each scaled amount to the nearest **0.05 ml** (the dispenser's smallest unit).
4. Each base has a **maximum tint %**: Pastel 2%, Medium 6%, Deep 12%. If total colorant (ml) is more than that % of the can volume, reject the request with a clear error message.
5. **Price** = base price per litre × litres + sum of (colorant ml × colorant cost per ml). Round to 2 decimals.

Worked example: a shade needs 3.2 ml Black per litre in Medium base. For a 4L can: $3.2 \times 4 = 12.8$ ml. Total colorant must be at most 6% of 4,000 ml, which is 240 ml, so the request is valid.

Requirements

Database

- Create the tables described in the table structure below
- EF Core migration that creates the schema
- Seed the data from the provided JSON file (see Appendix)

Table structure

Use these tables as the minimum design. You may add columns or indexes if you explain why in the README. All money and quantity columns use `decimal`, never `float`.

Bases

Column	Type	Rules
Id	int	Primary key, identity
Name	nvarchar(50)	Required, unique
MaxTintPercent	decimal(5,2)	Required, e.g. 6.00
PricePerLitre	decimal(10,2)	Required

Colorants

Column	Type	Rules
Id	int	Primary key, identity
Code	nvarchar(10)	Required, unique, e.g. C01
Name	nvarchar(50)	Required
CostPerMl	decimal(10,4)	Required

Shades

Column	Type	Rules
Id	int	Primary key, identity
Code	nvarchar(20)	Required, unique, e.g. OM-201
Name	nvarchar(100)	Required, indexed for search
HexColor	char(7)	Required, e.g. #7FA7B5

FormulaItems

Column	Type	Rules
Id	int	Primary key, identity

Column	Type	Rules
ShadeId	int	Foreign key to Shades
BaseId	int	Foreign key to Bases
ColorantId	int	Foreign key to Colorants
MLPerLitre	decimal(10,4)	Required, greater than 0

Unique constraint on (ShadeId, BaseId, ColorantId).

DispenseJobs

Column	Type	Rules
Id	int	Primary key, identity
ShadeId	int	Foreign key to Shades
BaseId	int	Foreign key to Bases
CanSizeLitres	decimal(5,2)	Required: 1, 4, 10 or 20
TotalColorantML	decimal(10,2)	Required
TintPercent	decimal(5,2)	Required
TotalPrice	decimal(12,2)	Required
CreatedAtUtc	datetime2	Required, default current UTC time

DispenseJobItems

Column	Type	Rules
Id	int	Primary key, identity
DispenseJobId	int	Foreign key to DispenseJobs, cascade delete
ColorantId	int	Foreign key to Colorants
DispensedML	decimal(10,2)	Required
Cost	decimal(10,2)	Required

DispenseJobItems stores what was actually dispensed, so past jobs stay correct even if a formula or price changes later.

Relationships: a Shade has many FormulaItems per Base; a DispenseJob has many DispenseJobItems.

Web API

Method	Endpoint	Purpose
GET	/api/shades?search=	List shades, filtered by name or code
GET	/api/shades/{id}	Shade details with its formula
POST	/api/tint/calculate	Input: shadeId, baseId, canSize. Returns scaled amounts, total tint % and price, or a validation error
POST	/api/dispense-jobs	Save a dispense job

- Return clear error responses (400 for invalid input, 404 for not found)
- Keep business logic out of the controllers

WPF client (one main screen, MVVM)

- Search box and a shade list showing a color swatch for each shade
- Dropdowns for base and can size
- A grid showing calculated colorant amounts, total tint % and price
- A **Dispense** button that saves the job and shows a confirmation
- Show API errors to the user; the UI must not freeze during API calls

UI design standards

The WPF screen must look like a professional commercial product, not a default Windows form. UI quality is part of the evaluation.

Screen layout

[Logo] Tint Studio		Store: Demo Store 01		
[Search shade...]	[Large swatch]	Ocean Mist OM-201		
[■] Ocean Mist OM-201	Base: [Medium v] Can size: [4L v]			
[■] Terracotta TR-115				
[■] ...				
	Colorant	Code	Amount (ml)	Cost
	Phthalo Blue	C03	17.40	20.88
	Black	C01	4.40	3.52
	Total colorant 21.80 ml			
	Tint used [#####-----] 0.55% of 6%			
	Price ₹1,104.40	[DISPENSE]		
● Connected to API		Last job: #1042		

Visual style

- Use a consistent theme defined in a ResourceDictionary: colors, fonts, font sizes, spacing and control styles. No hard-coded colors or fonts inside individual controls.
- A modern UI library is allowed and encouraged, such as MaterialDesignInXamlToolkit, WPF UI (wpfui) or ModernWpf. Custom styles are also fine.
- One primary accent color for main actions (the Dispense button), neutral colors elsewhere.
- Consistent spacing (e.g. multiples of 8 px), aligned labels and inputs, readable font (Segoe UI or similar, 13–14 px body).
- Each shade in the list shows a color swatch, and the selected shade shows a large swatch.

Layout and behaviour

- Minimum window size 1280 × 720; the layout resizes cleanly using Grid star sizing (no absolute positioning).
- Show a loading indicator during API calls and disable the Dispense button while busy.
- Show validation and API errors inline in a clear, friendly message (e.g. "Tint exceeds 6% limit for Medium base"), not raw exception text.
- Show success as a toast or snackbar notification with the job number.
- Format numbers consistently: ml to 2 decimals, prices as ₹ with thousand separators.
- Keyboard support: Enter runs search, logical Tab order, Dispense reachable without a mouse.
- A status bar shows API connection status.

Include 2–3 screenshots of the running app in the README.

Unit tests

- At least 5 tests covering scaling, rounding, the tint limit and price calculation

Scope limits

Not required: login or authentication, multi-tenancy, offline sync, reports, admin screens for editing data.

Optional, only if time is left (pick one):

- Dispense history screen
- Swagger documentation with request examples
- Docker setup for the API and database

Deliverables

- A Git repository link (with regular commits, not a single final commit)
- A README with setup steps, your assumptions and anything left unfinished
- Suggested plan: Day 1 for database, API and tests; Day 2 for the WPF screen and README

After submission, there will be a 20-minute call where you walk through your code and make one small change live.

Evaluation criteria

Area	Points
Business rules implemented correctly	25
API design and validation	20
WPF UI design and MVVM quality	20
Database design and EF Core usage	15
Unit tests	10
README and code readability	10
Total	100

Appendix: sample seed data

Use this format for the seed file. You may add more shades and colorants.

```
{
  "bases": [
    { "id": 1, "name": "Pastel", "maxTintPercent": 2, "pricePerLitre": 250 },
    { "id": 2, "name": "Medium", "maxTintPercent": 6, "pricePerLitre": 270 },
    { "id": 3, "name": "Deep", "maxTintPercent": 12, "pricePerLitre": 290 }
  ],
  "colorants": [
    { "id": 1, "code": "C01", "name": "Black", "costPerMl": 0.80 },
    { "id": 2, "code": "C02", "name": "Oxide Red", "costPerMl": 0.90 },
    { "id": 3, "code": "C03", "name": "Phthalo Blue", "costPerMl": 1.20 },
    { "id": 4, "code": "C04", "name": "Yellow Oxide", "costPerMl": 0.85 }
  ],
  "shades": [
    {
      "id": 1, "code": "OM-201", "name": "Ocean Mist", "hex": "#7FA7B5",
      "formulas": [
        { "baseId": 2, "items": [
          { "colorantId": 3, "mlPerLitre": 4.35 },
          { "colorantId": 1, "mlPerLitre": 1.10 }
        ] }
      ]
    },
    {
      "id": 2, "code": "TR-115", "name": "Terracotta", "hex": "#C4663F",
      "formulas": [
        { "baseId": 3, "items": [
          { "colorantId": 2, "mlPerLitre": 38.50 },
          { "colorantId": 4, "mlPerLitre": 22.20 },
          { "colorantId": 1, "mlPerLitre": 2.05 }
        ] }
      ]
    }
  ]
}
```

```
]
}
```

Prices are in SAR.